

ASSIGNMENT - 01

Question 1:

I would first recommend GenBank because it is one of the largest public databases for DNA and RNA sequences. The researcher can search the newly discovered DNA sequence to check whether it has already been reported by another scientist. If a similar sequence is found, it may also provide details such as the organism, gene name, and related research.

Next, I would use NCBI Gene to understand whether the sequence belongs to a known gene. This database provides information about gene function, chromosome location, gene structure, and links to scientific literature. It helps determine whether the sequence is associated with any known biological process or disease.

To study the relationship with the rare genetic disorder, I would also use OMIM (Online Mendelian Inheritance in Man). OMIM contains detailed information about human genes and inherited diseases. If the gene has already been linked to a genetic disorder, OMIM explains the disease, its symptoms, inheritance pattern, and known mutations.

Finally, I would refer to PubMed to find published research articles related to the sequence or disease. Reading previous studies can provide valuable evidence before performing laboratory experiments.

By combining these databases, the researcher can confirm whether the DNA sequence is already known, understand its function, and identify any possible connection with the rare genetic disorder.

Question 2:

To compare bacterial genomes, I would first recommend NCBI Genome because it contains complete genome sequences of many bacterial species. It allows researchers to compare genomes and study similarities and differences between organisms.

Another useful database is Ensembl Bacteria, which provides genome annotations, gene locations, and comparative genomics tools. It helps identify conserved genes that are shared among related bacterial species. Conserved genes are usually important because they perform essential biological functions and remain similar during evolution.

I would also use GenBank to obtain DNA sequences of different bacterial species. These sequences can be downloaded and compared using bioinformatics tools to identify conserved regions.

In addition, KEGG is helpful because it links genes with metabolic pathways and biological functions. After identifying conserved genes, KEGG can explain how those genes participate in important cellular processes or disease mechanisms.

Using these databases together gives a complete understanding of the bacterium. NCBI Genome and GenBank provide genome data, Ensembl Bacteria helps with comparison and annotation, and KEGG explains the biological role of the conserved genes. This information is valuable for understanding bacterial evolution and developing new treatments.

Question 3:

I would recommend UniProt as the first database because it is one of the most reliable protein databases. It contains protein sequences, names, functions, structures, and information about different organisms. By searching the isolated protein sequence, researchers can find similar proteins from other species and understand their possible functions.

I would also use Protein Data Bank (PDB) if structural information is available. PDB stores three-dimensional protein structures, which help scientists understand how proteins work and interact with other molecules.

Another useful database is Pfam, which identifies conserved protein domains and families. Even if the complete protein is new, the presence of known domains can provide clues about its function.

Finally, NCBI Protein is useful because it contains protein sequences submitted from different organisms. Researchers can compare their protein with sequences available in the database and identify evolutionary relationships.

Together, these databases provide information about protein sequence similarity, structure, function, conserved domains, and evolutionary relationships. This helps the biotechnology company understand whether the newly isolated protein has already been identified in other organisms and predict its biological role.

Question 4:

To determine the function of an unknown protein, I would first use UniProt because it provides detailed information about protein sequences, biological functions, cellular locations, and involvement in biological processes. If a similar protein exists, UniProt can give valuable functional information.

Next, I would use Pfam to identify conserved protein domains. Protein domains often indicate the function of a protein, even when the complete sequence has not been studied before.

If structural information is needed, I would use the Protein Data Bank (PDB). Proteins with similar structures often perform similar functions. Studying the three-dimensional structure can therefore help predict the protein's role.

I would also use KEGG to determine whether the protein participates in metabolic or signalling pathways. This provides insight into its biological importance.

Finally, PubMed can be used to read research papers describing similar proteins and experimental studies.

Using these databases together provides sequence information, conserved domains, structural details, biological pathways, and scientific evidence. Combining all this information gives a more accurate prediction of the protein's function before performing laboratory experiments.

Question:5

If I receive an unknown biological sequence, I would first examine the sequence itself. If it contains only the letters A, T, G, and C, it is most likely DNA. If it contains A, U, G, and C, it is probably RNA. If the sequence consists of the twenty amino acid letters, it is a protein sequence.

After identifying the possible sequence type, I would search GenBank for DNA or RNA sequences and UniProt for protein sequences. These databases help determine whether similar sequences have already been reported.

Next, I would compare the sequence with known sequences to identify similarities. A close match may reveal the sequence's identity, function, or the organism from which it originated.

If the sequence is identified as a gene, I would use NCBI Gene to study its function and gene information. If it is a protein, I would use Pfam to identify conserved domains and PDB to study any available protein structures.

Finally, I would use PubMed to review published research related to the sequence. This helps confirm the findings and provides additional biological information. Following this step-by-step approach allows accurate identification of the sequence and helps understand its biological significance using reliable biological databases.